

Clustering

Marcos Laserna Moratalla

2026-02-14

Librerías

```
rm(list = ls())

library(readxl)
library(xts)
library(zoo)
library(quantmod)
library(igraph)
library(ggraph)
library(shiny)
library(archetypes)
library(ggplot2)
library(tidyr)
library(dplyr)
library(biwavelet)
library(factoextra)
```

Preprocesado

```
# Importación desde Excel

procesar_activo_mensual <- function(archivo, nombre_hoja = 1, lineas_saltar = 0) {
  df <- as.data.frame(read_excel(archivo, sheet = nombre_hoja, skip = lineas_saltar))
  fechas <- as.Date(df[, 1])
  precios <- as.numeric(gsub(",", "", df[, 2]))
  serie <- na.omit(xts(precios, order.by = fechas))
  serie_mes <- serie[endpoints(serie, on = "months"), ]
  retornos <- na.omit(diff(log(serie_mes)))
  index(retornos) <- as.yearmon(index(retornos))
  return(retornos)
}

# Índices de Renta Variable Verde

csi <- procesar_activo_mensual("CSI_New_Energy_index.xlsx", 1, 29)
wind <- procesar_activo_mensual("ISE_Clean_Edge_Global_Wind_Energy_index.xlsx", 1, 18)
```

```

sp      <- procesar_activo_mensual("S&P_Global_Clean_energy_index.xlsx", 1, 17)
wild   <- procesar_activo_mensual("WilderHill.xlsx", 1, 18)
sun     <- procesar_activo_mensual("DATA_revisados_Hui.xlsx", "SUNIDX", 0)
uranio <- procesar_activo_mensual("DATA_revisados_Hui.xlsx", "MVIS_GLO_URAN_PR", 0)
ftse    <- procesar_activo_mensual("DATA_revisados_Hui.xlsx", "FTSE_ENV_OPPORT_ENE_EFF", 0)
erixp   <- procesar_activo_mensual("DATA_revisados_Hui.xlsx", "ERIXP_USD", 0)

rv_unida <- merge.xts(csi, wind, sp, wild, sun, uranio, ftse, erixp, join = "inner")
colnames(rv_unida) <- c("CSI_China", "Wind", "SP_Clean", "WilderHill", "Solar", "Uranium",
                        "FTSE_Env", "ERIXP_EU")

# Bonos Países (sin LT)
df_bonos <- as.data.frame(read_excel("eurostat_bonos10.xlsx", sheet = "Bond_2015"))
fechas_bonos <- as.yearmon(df_bonos[, 1])
columnas_a_quitar <- c("Time", "LT")
bonos_sel <- df_bonos[, !(names(df_bonos) %in% columnas_a_quitar)]

# Log - Retornos
bonos_gross <- 1 + (as.matrix(bonos_sel) / 100)
bonos_diff <- diff(log(bonos_gross))
bonos_xts <- xts(bonos_diff, order.by = fechas_bonos[-1])

# ETF Bonos Verdes
getSymbols("BGRN", src = "yahoo", from = "2015-01-01", auto.assign = TRUE)

## [1] "BGRN"

bgrn_m <- BGRN[endpoints(BGRN, on = "months"), ]
ret_bgrn <- na.omit(diff(log(Ad(bgrn_m))))

colnames(ret_bgrn) <- "GR_BOND"
index(ret_bgrn) <- as.yearmon(index(ret_bgrn))

dataset_master <- merge.xts(rv_unida, bonos_xts, ret_bgrn, join = "inner")

```

Gráficas de los Bonos

```

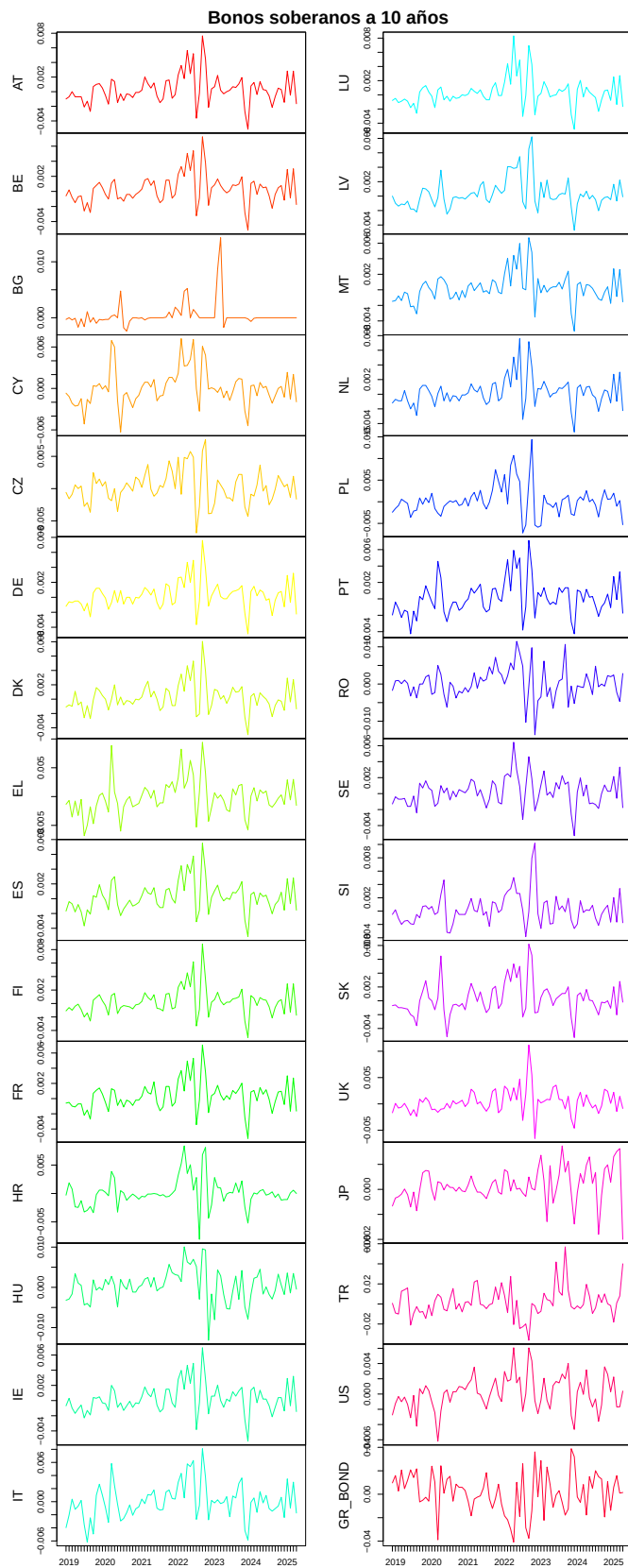
# Graficamos bonos normales
par(mar = c(2, 4, 2, 1) + 0.1, oma = c(4, 0, 3, 0))

num_bonos <- ncol(dataset_master) - 8
colores_bonos <- rainbow(num_bonos)

plot.zoo(dataset_master[, 9:ncol(dataset_master)],
          col = colores_bonos,
          lwd = 1,
          ylab = colnames(dataset_master)[9:ncol(dataset_master)],
          xlab = "", main = "")

mtext("Bonos soberanos a 10 años", side = 3, outer = TRUE, cex = 1.3, font = 2)

```



Dendograma

```
# Datos
data_mat <- scale(as.matrix(dataset_master))
fin <- colnames(data_mat)
n <- ncol(data_mat)

# Convertimos en matriz
tiempo <- 1:nrow(data_mat)

# Aquí guardaremos los coeficientes R2
adj_R2 <- matrix(0, n, n, dimnames = list(fin, fin))

# Calculamos la coherencia Wavelet
for(i in 2:n) {
  for(j in 1:(i-1)) {
    d1 <- cbind(tiempo, data_mat[, i])
    d2 <- cbind(tiempo, data_mat[, j])

    wtc_res <- wtc(d1, d2, nrands = 0)

    # Filtramos por escala
    idx_band <- which(wtc_res$period > 12)

    # Extraemos el R^2
    mean_r2 <- mean(wtc_res$rsq[idx_band, ], na.rm = TRUE)

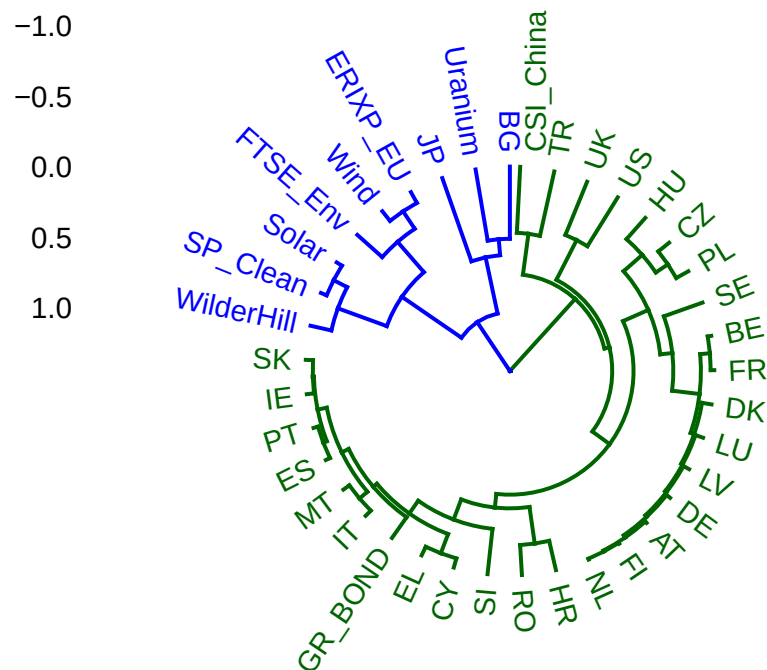
    # Guardamos el valor
    adj_R2[i,j] <- adj_R2[j,i] <- mean_r2
  }
}
diag(adj_R2) <- 1

# 3. Calcular la distancia y clusterizamos
dist_matrix <- as.dist(1 - adj_R2)

# Agrupamos por de Ward
hc_macro <- hclust(dist_matrix, method = "ward.D2")

# Visualizamos
k_clusters <- 2
colores_clusters <- c("darkgreen", "blue")

fviz_dend(hc_macro, k = k_clusters, cex = 0.8,
          k_colors = colores_clusters,
          type = "circular", rect = TRUE,
          main = "Dendrograma") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold")) +
  theme(legend.position = "none")
```



Arquetipos en Europa

```
# Matriz de distancias
matriz_distancia_total <- 1 - adj_R2

# Seleccionamos los bonos de países alrededor de Europa
indices_rv <- c("CSI_China", "Wind", "SP_Clean", "WilderHill", "Solar", "Uranium",
               "FTSE_Env", "ERIXP_EU")
bonos_verdes_etf <- "GREEN_BOND_GLOBAL"
extranjeros <- c("JP", "US", "TR", "UK")

países_soberanos <- setdiff(colnames(adj_R2), c(indices_rv, bonos_verdes_etf, extranjeros))

matriz_final_dist <- matriz_distancia_total[países_soberanos, países_soberanos]

# Modelo de 3 arquetipos
set.seed(33)

modelo_dist <- stepArchetypes(data = matriz_final_dist, k = 3, verbose = FALSE)
mejor_dist <- bestModel(modelo_dist)

# Coeficiente para elegir extremos
df_alphas <- as.data.frame(coef(mejor_dist, "alphas"))
```

```

df_alphas$Pais <- rownames(matriz_final_dist)

# fijamos FI y BG como estables y aislado
idx_central <- which.max(df_alphas[which(df_alphas$Pais == "FI"), 1:3])
idx_aislado <- which.max(df_alphas[which(df_alphas$Pais == "BG"), 1:3])
idx_mixto <- setdiff(1:3, c(idx_central, idx_aislado))

column_names <- character(3)
column_names[idx_central] <- "Central"
column_names[idx_aislado] <- "Aislado"
column_names[idx_mixto] <- "Mixto"
colnames(df_alphas)[1:3] <- column_names

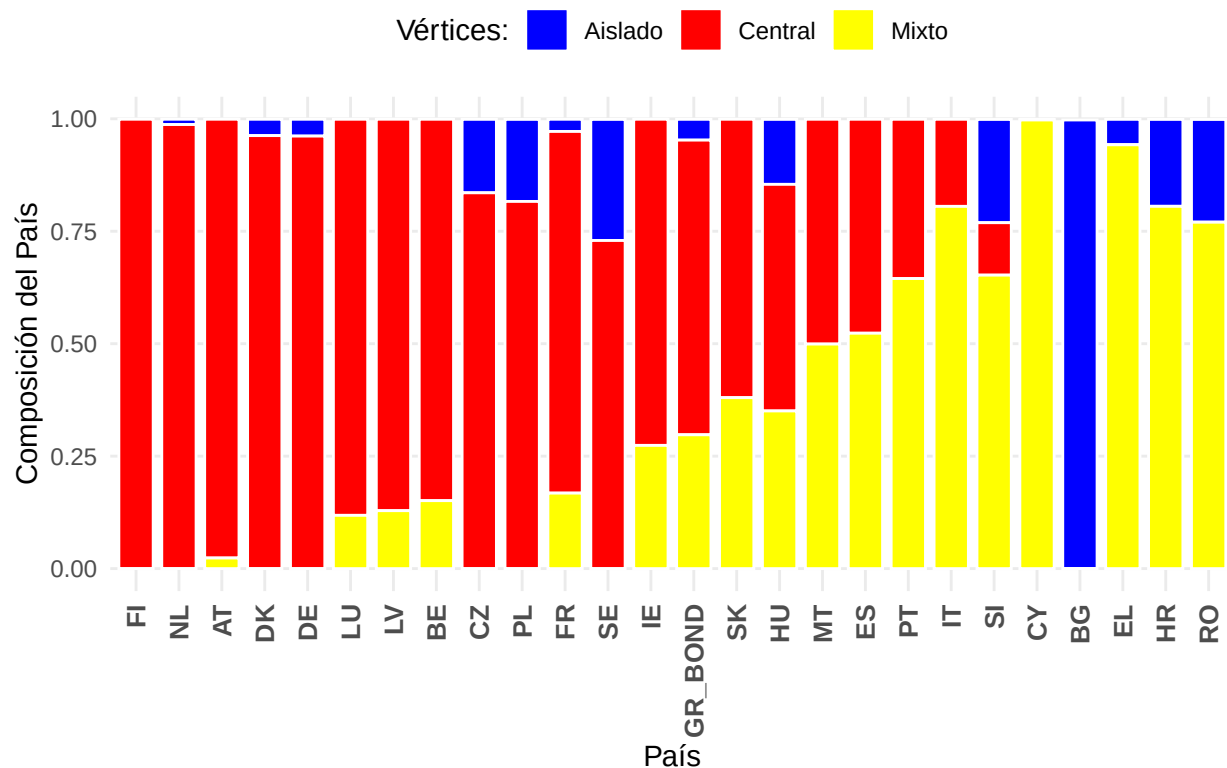
# Visualizamos
df_plot <- pivot_longer(df_alphas,
                        cols = c("Aislado", "Central", "Mixto"),
                        names_to = "Perfil_Geometrico",
                        values_to = "Peso")

orden_paises <- df_alphas$Pais[order(-df_alphas$Central)]
df_plot$Pais <- factor(df_plot$Pais, levels = orden_paises)

# Graficamos
ggplot(df_plot, aes(x = Pais, y = Peso, fill = Perfil_Geometrico)) +
  geom_bar(stat = "identity", width = 0.8, color = "white") +
  scale_fill_manual(values = c("Aislado" = "blue",
                              "Central" = "red",
                              "Mixto" = "yellow"),
                  labels = c("Aislado",
                              "Central",
                              "Mixto")) +
  theme_minimal() +
  labs(title = "Europa",
       x = "País",
       y = "Composición del País",
       fill = "Vértices:") +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1,
                                    face = "bold", size = 10),
        legend.position = "top",
        plot.title = element_text(face = "bold", size = 16),
        panel.grid.minor = element_blank())

```

Europa



Arquetipos en Eurozona

```
# Matriz de distancias
matriz_distancia_total <- 1 - adj_R2

# Solo Eurozona
eurozona_codigos <- c("AT", "BE", "CY", "EE", "FI", "FR", "DE", "GR",
                      "IE", "IT", "LV", "LT", "LU", "MT", "NL", "PT",
                      "SK", "SI", "ES")

países_soberanos <- intersect(colnames(adj_R2), eurozona_codigos)
matriz_final_dist <- matriz_distancia_total[países_soberanos, países_soberanos]

# 3 arquetipos
set.seed(33)
modelo_dist <- stepArchetypes(data = matriz_final_dist, k = 3, verbose = FALSE)
mejor_dist <- bestModel(modelo_dist)

# Coeficientes de extremos
df_alphas <- as.data.frame(coef(mejor_dist, "alphas"))
df_alphas$País <- rownames(matriz_final_dist)

# Fijamos NL como estable
idx_central <- which.max(df_alphas[df_alphas$País == "NL", 1:3])

# Calculamos el resto por varianza
restantes <- setdiff(1:3, idx_central)
```

```

reps_nombres <- paises_soberanos[apply(coef(mejor_dist, "alphas"), 2, which.max)]
dist_medias <- colMeans(matriz_final_dist)

nombres_cols <- character(3)
nombres_cols[idx_central] <- "Central"

if(dist_medias[reps_nombres[restantes[1]]] > dist_medias[reps_nombres[restantes[2]]]) {
  nombres_cols[restantes[1]] <- "Aislado"
  nombres_cols[restantes[2]] <- "Mixto"
} else {
  nombres_cols[restantes[1]] <- "Mixto"
  nombres_cols[restantes[2]] <- "Aislado"
}

colnames(df_alphas)[1:3] <- nombres_cols

#Visualizamos
df_plot <- pivot_longer(df_alphas,
                        cols = c("Aislado", "Central", "Mixto"),
                        names_to = "Perfil_Geometrico",
                        values_to = "Peso")

orden_paises <- df_alphas$Pais[order(-df_alphas$Central)]
df_plot$Pais <- factor(df_plot$Pais, levels = orden_paises)

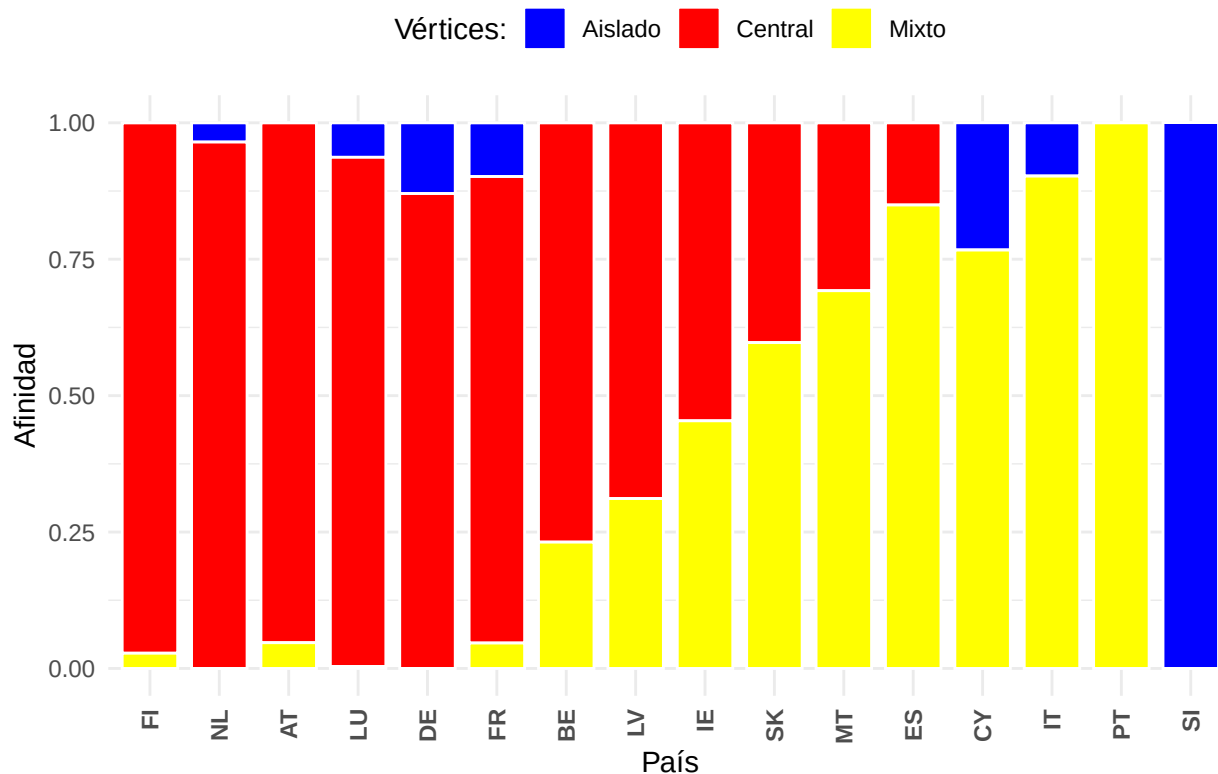
# Graficamos

ggplot(df_plot, aes(x = Pais, y = Peso, fill = Perfil_Geometrico)) +
  geom_bar(stat = "identity", width = 0.8, color = "white") +
  scale_fill_manual(values = c("Aislado" = "blue",
                              "Central" = "red",
                              "Mixto" = "yellow")) +

  theme_minimal() +
  labs(title = "Eurozona",
       x = "País", y = "Afinidad", fill = "Vértices:") +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1, face = "bold"),
        legend.position = "top")

```


Eurozona



Arquetipos en Europa (Dinámico)

```
ui <- fluidPage(
  titlePanel("Arquetipos Wavelet"),

  sidebarLayout(
    sidebarPanel(

      sliderInput("rango", "Frecuencia:",
        min = 1, max = 64, value = c(16, 32), step = 1),

      hr(),
      tags$b("Leyenda de Colores:"),
      tags$p(style="color:#8B0000;", "Central"),
      tags$p(style="color:#104E8B;", "Aislado"),
      tags$p(style="color:#E69F00;", "Mixto"),
      hr()
    ),

    mainPanel(
      plotOutput("graficoArquetipos", height = "650px")
    )
  )
)
```

```

)

server <- function(input, output) {

matriz_reactiva <- reactive({
  total_cols <- ncol(dataset_master)
  nombres_soberanos <- colnames(dataset_master)[9:(total_cols - 1)]
  paises_excluir <- c("JP", "TR", "US", "UK", "LT")
  soberanos_finales <- nombres_soberanos[!(nombres_soberanos %in% paises_excluir)]

  # Usamos la matriz de Coherencia Wavelet
  m_wavelet <- adj_R2[soberanos_finales, soberanos_finales]
  frec_media <- mean(input$rancho)

  dist_dinamica <- 1 - (m_wavelet^(1 + (frec_media / 100)))
  return(dist_dinamica)
})

output$graficoArquetipos <- renderPlot({
  m_dist <- matriz_reactiva()

  # Modelamos arquetipos
  set.seed(33)
  modelo <- stepArchetypes(data = m_dist, k = 3, verbose = FALSE)
  mejor <- bestModel(modelo)

  # Coeficientes extremos
  df_alphas <- as.data.frame(coef(mejor, "alphas"))
  paises <- rownames(m_dist)
  rownames(df_alphas) <- paises

  # Prefijamos dos extremos
  idx_central <- which.max(df_alphas["FI", ])
  idx_aislado <- which.max(df_alphas["BG", ])
  idx_mixto <- setdiff(1:3, c(idx_central, idx_aislado))

  # Renombramos las columnas
  column_names <- character(3)
  column_names[idx_central] <- "Central"
  column_names[idx_aislado] <- "Aislado"
  column_names[idx_mixto] <- "Mixto"
  colnames(df_alphas)[1:3] <- column_names

  df_alphas$Pais <- paises

  df_alphas <- df_alphas[order(-df_alphas$Central), ]
  df_alphas$Pais <- factor(df_alphas$Pais, levels = df_alphas$Pais)

  df_plot <- pivot_longer(df_alphas,
    cols = c("Central", "Aislado", "Mixto"),
    names_to = "Arquetipo",
    values_to = "Peso")

```

```

# Graficamos
ggplot(df_plot, aes(x = Pais, y = Peso, fill = Arquetipo)) +
  geom_bar(stat = "identity", width = 0.8, color = "white") +
  scale_fill_manual(values = c("Central" = "#8B0000",
                                "Aislado" = "#104E8B",
                                "Mixto" = "#E69F00"),
                    labels = c("Aislado", "Central", "Mixto")) +
  theme_minimal() +
  labs(title = "Arquetipos Wavelet",
        subtitle = paste("Escala Wavelet:",
                          input$rancho[1], "-", input$rancho[2], "| d = 1 - R2"),
        x = "País",
        y = "Alphas",
        fill = "Extremos:") +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust = 1, face = "bold",
                                     size = 11),
        legend.position = "top",
        plot.title = element_text(face = "bold", size = 18),
        panel.grid.minor = element_blank())
})
}

shinyApp(ui = ui, server = server)

```

Gráfico general (Dinámico)

```

ui <- fluidPage(
  titlePanel("Correlación Wavelet: Bonos Soberanos e Índices Verdes"),
  sidebarLayout(
    sidebarPanel(
      sliderInput("rancho_frecuencia", "Frecuencia:", min = 1, max = 64,
                  value = c(16, 32), step = 1, round = TRUE),
      sliderInput("umbral", "Umbral de Conexión:", min = 0.1, max = 0.9,
                  value = 0.42, step = 0.05)
    ),
    mainPanel(
      plotOutput("redPlot", height = "650px")
    )
  )
)

server <- function(input, output) {

  output$redPlot <- renderPlot({

    if(!exists("adj_R2")) {
      return(ggplot() + annotate("text", x=0, y=0, label="ERROR: matriz.
                                ", size=5, color="red") + theme_void())
    }
  })
}

```

```

# Matriz
factor_escalas <- input$rancho_frecuencia[2] / 64
matriz_dinamica <- adj_R2 * factor_escalas
diag(matriz_dinamica) <- 0

n <- ncol(matriz_dinamica)
fin <- colnames(matriz_dinamica)

# Filtramos por umbral
edges <- data.frame(from=character(), to=character(), weight=numeric())
for(i in 1:(n-1)) {
  for(j in (i+1):n) {
    w <- matriz_dinamica[i,j]
    if(!is.na(w) && w > input$umbral) {
      edges <- rbind(edges, data.frame(from=fin[i], to=fin[j], weight=w))
    }
  }
}

if(nrow(edges) == 0) {
  return(ggplot() + annotate("text", x=0, y=0, label="El umbral es muy alto.
                                     No hay conexiones.", size=6, color="red") + theme_void())
}

# Graficamos
net <- graph_from_data_frame(edges, vertices=fin, directed=FALSE)

indices_rv <- c("CSI_China", "Wind", "SP_Clean", "WilderHill", "Solar",
               "Uranium", "FTSE_Env", "ERIXP_EU")
bonos_verdes_tecnicos <- c("GRNENEF", "GRNFUEL", "GRNGB", "GRNPOL", "GRNSOLAR",
                           "GRNTRN", "GRNWIND")

V(net)$Sector <- ifelse(V(net)$name %in% indices_rv, "Índices Verdes",
                       ifelse(V(net)$name %in% bonos_verdes_tecnicos, "Bonos Verdes",
                              "Bonos Soberanos"))

V(net)$Degree <- strength(net)

set.seed(555)

# Visualizamos
ggraph(net, layout = "fr") +
  geom_edge_link(aes(width = weight, alpha = weight), color = "gray50") +
  scale_edge_width(range = c(0.8, 3), name="Fuerza R2") +
  scale_edge_alpha(range = c(0.2, 0.8), guide="none") +

  geom_node_point(aes(color = Sector, size = Degree)) +
  scale_size(range = c(5, 15), guide = "none") +

  scale_color_manual(values = c("Bonos Soberanos" = "#FFA500",
                                "Índices Verdes" = "#32CD32",
                                "Bonos Verdes" = "#006400")) +

```

```

    geom_node_text(aes(label = name), repel = TRUE, fontface="bold", size=4.5,
                  bg.color="white", bg.r=0.15) +
    theme_graph() +
    theme(legend.position = "right",
          legend.title = element_text(face="bold", size=12)) +
    labs(title = "Red de Coherencia",
         subtitle = paste("Umbral de Coherencia Wavelet R2 >", input$umbral))
  })
}

shinyApp(ui = ui, server = server)

```

K-Means

```

library(factoextra)
library(cluster)
library(NbClust)

# Matriz
matriz_completa_dist <- 1 - adj_R2

# Óptimo k con Silhouette Index
set.seed(33)
stats_clusters <- fviz_nbclust(matriz_completa_dist, kmeans, method = "silhouette")

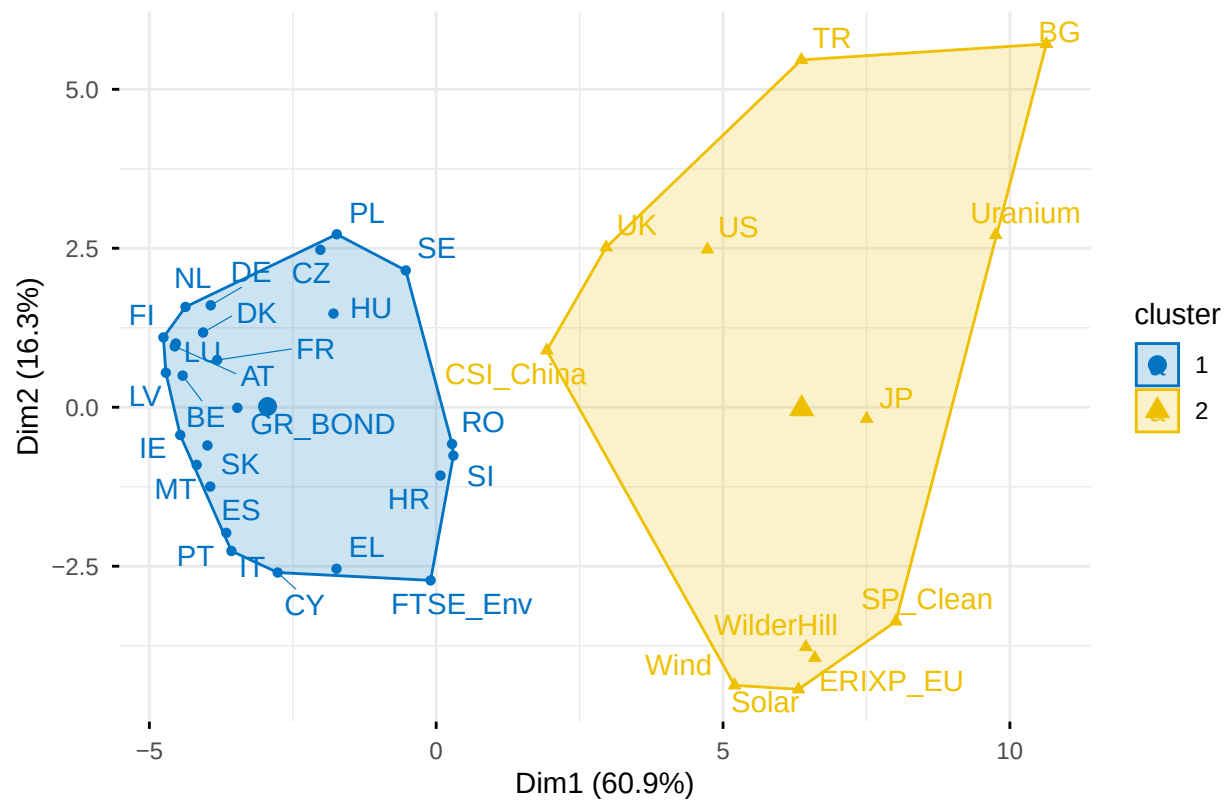
k_autodetectado <- as.numeric(stats_clusters$data$clusters[which.max(stats_clusters$data$y)])

# k-means
res_km <- kmeans(matriz_completa_dist, centers = k_autodetectado, nstart = 25)

# Visualización del PCA
fviz_cluster(res_km, data = matriz_completa_dist,
             palette = "jco",
             geom = c("point", "text"),
             ellipse.type = "convex",
             repel = TRUE,
             main = paste("Clustering Automático (k =", k_autodetectado, ")"),
             ggtheme = theme_minimal())

```

Clustering Automático (k = 2)



```
lista_grupos <- data.frame(
  Activo = names(res_km$cluster),
  Cluster = res_km$cluster
) %>% arrange(Cluster)
```